

Ten Percent

What the 10% rule really is, why the top of a pyramid moves so much faster than the bottom, and which kind of pyramid is allowed to be upside down.

Name _____

Date _____

Period _____

Demonstration: andresdbp.github.io/resources/demos/ten-percent/

1 Before you open anything

Answer from what you already know. Being wrong here costs nothing — the point is to have a number on paper that the demonstration can settle.

PREDICTION

A food chain has **5 trophic levels**. The producers fix **10,000 kJ m⁻² yr⁻¹**. Transfer efficiency is raised from **10%** to **20%** — doubled. By what factor does the energy reaching the *top* level change?

Write one sentence saying how you got that factor.

2 Reading the pyramid

Open the demonstration. Leave the **Energy per year** tab selected and press **Reset to 10%** so everyone starts from the same place: 10.0% efficiency, 10,000 producer energy, 5 levels.

1. Confirm the three control readouts say **10.0%**, **10,000** and **5 levels**.
2. Copy the **Energy** and **Share of producers** columns out of the level table in the **Readouts** panel.
3. Then record the three boxes at the top of that panel.

Level	Energy ($\text{kJ m}^{-2} \text{yr}^{-1}$)	Share of producers
1. Producers		
2. Herbivores		
3. Carnivores		
4. Top carnivores		
5. Apex predators		

Top level	Predators	Base : top

a. Going down the *Share of producers* column, each entry is the one above it multiplied by what?

b. There are 5 levels but only *four* transfers between them. Write the energy at level 5 as producer energy multiplied by a power of the efficiency, and say why the exponent is 4 and not 5.

c. Read the note printed under the *Predators* box. It gives a density in animals per km^2 and the range real 20 kg carnivores actually live at. Copy both.

3 Which control changes the shape

Two sliders feed the same pyramid. They do not do the same job to it. Keep **5 levels** selected throughout this part.

1. Leave **Producer energy (NPP)** at **10,000**.
2. Set **Transfer efficiency** to each value in the table (the slider steps in 0.5% increments, so all six are reachable exactly) and record the three readout boxes.

Efficiency	Top level ($\text{kJ m}^{-2} \text{yr}^{-1}$)	Predators	Base : top
2.0%			
5.0%			
10.0%			
15.0%			
20.0%			
25.0%			

1. Now put **Transfer efficiency** back to **10.0%** and vary **Producer energy (NPP)** instead.

Producer energy	Top level ($\text{kJ m}^{-2} \text{yr}^{-1}$)	Predators	Base : top
2,000			
10,000			
20,000			
40,000			

a. One of the two sliders leaves *Base : top* completely unchanged. Which one, and why does that ratio not care about it?

b. In the first table, efficiency goes from 10.0% to 20.0%. Divide the two *Top level* values. Now do the same for 5.0% and 20.0%. State the rule in your own words: multiplying the efficiency by k across t transfers multiplies the top level by _____.

c. Go back to your prediction in part 1. Compare it with the 10.0% $\rightarrow$ 20.0% factor you just computed. If they differ, name the specific assumption that produced your original number.

d. Watch the *Producers* row of the level table during each experiment. In which one does it stay fixed while the top level swings by orders of magnitude? What does that tell you about where along the chain the efficiency slider does its work?

4 Using it the way a question would

Two things a test asks for: reading the model backwards, and knowing when a familiar shortcut stops holding.

1. Set **Producer energy (NPP)** to **10,000** and the chain to **5** levels.
 2. Working the slider, find the *lowest* efficiency setting at which the **Predators** box reaches at least **200**.
- a. Efficiency required: _____. Now set efficiency *one step lower* (0.5% less) and record the head count there: _____. One half of one percent of efficiency is worth how many predators?

1. Set **Transfer efficiency** to **2.0%**, its minimum, with 5 levels and NPP still at 10,000. The *Predators* box changes colour and the note under it changes wording.
2. Push **Producer energy (NPP)** all the way to its maximum, **40,000**, without touching anything else.
3. Leave NPP at 40,000 and switch the chain to **4** levels, then to **3**.

Setting (efficiency 2.0% throughout)	Top level	Predators
NPP 10,000, 5 levels		
NPP 40,000, 5 levels		
NPP 40,000, 4 levels		
NPP 40,000, 3 levels		

b. Quadrupling the producer energy is a large change to the base of the pyramid. At 2.0% and 5 levels, did it get the reserve to one whole predator? Removing a single link did. Explain, using the exponent, why shortening the chain beats enriching the base.

Now the shortcut that fails. Click the **Ocean plankton** tab.

	Phytoplankton	Zooplankton
Standing biomass (g m^{-2})		
Replaced about every		
Production ($\text{g m}^{-2} \text{ yr}^{-1}$)		

c. Record the two boxes below the table: $\text{Zoo} \div \text{phyto mass} = \text{-----}$ and $\text{Phyto} \rightarrow \text{zoo} = \text{-----}$. Show that the second one follows from the two production figures.

d. The shortcut "the bigger bar holds more energy" gives the wrong answer here. Using the turnover times, explain in two sentences how the zooplankton can carry five times the standing mass while receiving less than a quarter of the annual production.

The standing-crop figures are the classic English Channel values and the turnover times are rounded approximations that shift with season and species. Treat the production row as an order-of-magnitude estimate rather than a measurement, and do not quote it to three figures in an answer.

5 Solve these without the demonstration

Close the page or look away. Show your working. Then reopen it, check yourself, and if a number is off, write down exactly where.

1. Producers fix $20,000 \text{ kJ m}^{-2} \text{ yr}^{-1}$. The chain has 4 trophic levels and transfer efficiency is 10% at every step. How much energy reaches the top level, and what share of the producers' energy is that?

2. A 5-level chain starts from $10,000 \text{ kJ m}^{-2} \text{ yr}^{-1}$ of production and delivers $16 \text{ kJ m}^{-2} \text{ yr}^{-1}$ to the top level. Find the transfer efficiency, assuming it is the same at every step.

3. A 100 km^2 reserve ($1 \times 10^8 \text{ m}^2$) has 5 trophic levels, 10% efficiency and $10,000 \text{ kJ m}^{-2} \text{ yr}^{-1}$ of net primary production. A 20 kg carnivore needs about $2.53 \times 10^6 \text{ kJ}$ per year. How many can the reserve support? Watch your units.

4. No arithmetic for this one. A survey of a lake reports more zooplankton biomass than phytoplankton biomass, and a separate survey of the same lake reports annual phytoplankton production far above annual zooplankton production. Explain why both reports can be correct at once, and state which of the two quantities an energy pyramid is built from and why that one can never invert.

6 On your own

Nothing here is graded, and the demonstration does more than this worksheet uses.

- Watch the *Why the exponential is the whole lesson* panel while you switch between 3, 4 and 5 levels. The multiplier printed there is 4 raised to the number of transfers. Work out why the base is 4 and not 15, then argue from that number why real food chains rarely run past four or five links.
- The note under *Predators* reports a density per km^2 and compares it with the 0.01–0.5 per km^2 that real 20 kg carnivores live at. Find combinations of efficiency, NPP and chain length that

land inside that realistic range, and see how narrow the window is.

- The hint under *Producer energy (NPP)* names a biome and converts the value to grams of dry plant matter per m² per year at 18 kJ g⁻¹. Set it to a desert value and a rainforest value and decide how much of the difference between those two ecosystems' top predators is production and how much is efficiency.